MZ UCA Cloud Academy

AWS Solutions Architect Associate (SAA)

Learning Notes 📝

AWS Organizations, Service Control Policies (SCPs), Landing Zones & Multi-Account Strategy

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◈ INTRODUCTION

As organizations grow in the cloud, managing everything inside a single AWS account becomes increasingly difficult.

A startup with three employees may successfully operate from one AWS account.

However, a large enterprise with hundreds or thousands of employees requires:

✔ Strong governance

✔ Security controls

✔ Cost visibility

✔ Compliance management

✔ Operational separation

✔ Risk isolation

AWS provides several services and architectural patterns to address these challenges, including:

◆ AWS Organizations

◆ Organizational Units (OUs)

◆ Service Control Policies (SCPs)

◆ Landing Zones

◆ Multi-Account Strategies

These services form the foundation of enterprise cloud governance.

◈ THE PROBLEM WITH A SINGLE AWS ACCOUNT

Many organizations begin with a single AWS account.

While simple initially, this model introduces challenges as the organization grows.

Common Problems

Security Risk

All resources share the same environment.

A security incident may affect everything.

Cost Visibility

Tracking spending across teams becomes difficult.

Compliance Challenges

Different departments often require different controls.

Operational Complexity

Production and development workloads become mixed together.

Increased Blast Radius

One mistake can impact multiple business systems.

Architecture Insight

As cloud adoption grows, organizations must separate workloads and implement governance controls.

◈ WHAT IS AWS ORGANIZATIONS?

AWS Organizations is a service that enables centralized management of multiple AWS accounts.

It allows organizations to create, organize, govern, and secure AWS accounts from a single management account.

Key Capabilities

✔ Centralized Account Management

✔ Consolidated Billing

✔ Governance Controls

✔ Security Standardization

✔ Policy Enforcement

✔ Organizational Structure

Architecture View

Instead of managing:

Account A

Account B

Account C

Individually,

AWS Organizations provides one central governance layer.

Benefits

✔ Simplified Administration

✔ Better Governance

✔ Improved Security

✔ Centralized Visibility

✔ Enterprise Scalability

◈ AWS ORGANIZATION STRUCTURE

An AWS Organization consists of:

Management Account

The primary account responsible for governance.

Member Accounts

Accounts that belong to the organization.

Organizational Units (OUs)

Logical containers used to group accounts.

Example

AWS Organization

│

├── Production OU

│ ├── Production Account

│ └── Customer Services Account

│

├── Development OU

│ ├── Development Account

│ └── Testing Account

│

├── Security OU

│ ├── Security Account

│ └── Log Archive Account

│

└── Sandbox OU

```
└── Student Account
```

Exam Tip

Organizational Units (OUs) help apply governance policies to multiple accounts simultaneously.

◈ ORGANIZATIONAL UNITS (OUs)

An Organizational Unit (OU) is a logical grouping of AWS accounts.

OUs help organizations organize cloud resources according to business needs.

Common OUs

Production OU

Hosts business-critical workloads.

Development OU

Hosts development environments.

Security OU

Hosts security tools and monitoring systems.

Sandbox OU

Hosts experimentation and learning environments.

Shared Services OU

Hosts centralized enterprise services.

Benefits

✔ Simplified Governance

✔ Policy Inheritance

✔ Easier Administration

✔ Better Structure

Architecture Principle

Design OUs around governance requirements rather than technical services.

◈ CONSOLIDATED BILLING

One major advantage of AWS Organizations is Consolidated Billing.

What is Consolidated Billing?

AWS combines billing information from all member accounts into one invoice.

Benefits

✔ Centralized Cost Management

✔ Easier Financial Reporting

✔ Simplified Accounting

✔ Cost Visibility

Example

Development Account

$500

Production Account

$2,000

Security Account

$300

Organization Total

$2,800

One invoice.

One payment process.

◈ SERVICE CONTROL POLICIES (SCPs)

Service Control Policies (SCPs) are governance policies used within AWS Organizations.

Definition

SCPs define the maximum permissions available within AWS accounts.

Critical Concept

SCPs do NOT grant permissions.

SCPs only limit permissions.

Think of SCPs as Guardrails

IAM Policy

↓

What users can do.

SCP

↓

What users are never allowed to do.

Architecture Insight

IAM grants access.

SCPs restrict access.

Both work together.

◈ HOW SCPs WORK

AWS evaluates:

IAM Permissions

AND

Service Control Policies

before allowing actions.

Example

IAM Policy

Allows:

ec2:TerminateInstances

SCP

Denies:

ec2:TerminateInstances

Result

❌ Access Denied

Exam Rule

Explicit Deny Always Wins

This rule applies throughout AWS security architecture.

◈ COMMON SCP USE CASES

Organizations frequently use SCPs to prevent dangerous actions.

Examples

Prevent deletion of CloudTrail

Prevent disabling GuardDuty

Prevent account closure

Prevent modification of security services

Prevent creation of resources in unauthorized Regions

Prevent use of restricted services

Benefits

✔ Improved Governance

✔ Reduced Risk

✔ Better Compliance

✔ Stronger Security Controls

◈ WHAT IS A LANDING ZONE?

A Landing Zone is a secure, scalable, pre-configured AWS environment.

It serves as the foundation for enterprise cloud adoption.

Purpose

Establish governance before workloads are deployed.

Includes

✔ Account Structure

✔ Security Controls

✔ Logging

✔ Networking

✔ IAM Standards

✔ Compliance Controls

✔ Monitoring

Architecture Principle

Build the foundation first.

Deploy workloads second.

◈ WHY LANDING ZONES MATTER

Without a Landing Zone:

❌ Inconsistent Configuration

❌ Security Gaps

❌ Governance Problems

❌ Compliance Risks

❌ Operational Complexity

With a Landing Zone:

✔ Consistent Standards

✔ Security Baselines

✔ Compliance Readiness

✔ Operational Efficiency

✔ Governance by Design

Business Benefit

Landing Zones accelerate cloud adoption while maintaining control.

◈ MULTI-ACCOUNT STRATEGY

A Multi-Account Strategy is the practice of separating workloads into different AWS accounts.

This is considered an AWS best practice.

Typical Account Structure

Production Account

Hosts live customer-facing systems.

Development Account

Hosts developer environments.

Testing Account

Hosts QA and testing workloads.

Security Account

Hosts security services.

Log Archive Account

Stores centralized logs.

Shared Services Account

Hosts shared enterprise services.

Sandbox Account

Hosts experimentation environments.

◈ BENEFITS OF MULTI-ACCOUNT ARCHITECTURE

Security Isolation

Compromise in one account does not automatically affect others.

Governance

Policies can be applied independently.

Cost Management

Spending is easier to track.

Compliance

Different workloads can meet different compliance requirements.

Operational Efficiency

Teams can work independently.

◈ THE BLAST RADIUS PRINCIPLE

One of the most important cloud architecture concepts is Blast Radius.

Definition

Blast Radius refers to the scope of impact when something goes wrong.

Example

Single Account

Developer accidentally deletes resources.

Impact:

Entire environment affected.

Multi-Account Architecture

Developer affects Development Account only.

Production remains protected.

Architecture Principle

Good architecture limits the blast radius of failure.

◈ ENTERPRISE GOVERNANCE MODEL

A mature AWS environment commonly follows this model:

AWS Organization

│

├── Security Account

├── Log Archive Account

├── Shared Services Account

├── Production Account

├──── Development Account

├──── Testing Account

└──── Sandbox Account

Protected by:

✔ SCPs

✔ IAM Controls

✔ CloudTrail

✔ GuardDuty

✔ Security Hub

✔ Compliance Monitoring

◈ AWS CERTIFICATION EXAM TIPS

Understand the following concepts thoroughly:

✔ AWS Organizations

✔ Organizational Units

✔ Consolidated Billing

✔ Service Control Policies

✔ Landing Zones

✔ Multi-Account Strategy

✔ Governance Controls

✔ Account Isolation

✔ Blast Radius

Frequently Tested

SCPs vs IAM Policies

IAM Policies:

Grant Permissions

SCPs:

Restrict Permissions

Remember:

SCPs Never Grant Permissions

◈ THINK LIKE A CLOUD GOVERNANCE ARCHITECT

Before creating an AWS account, ask:

◆ Why does this account exist?

◆ What workloads will run here?

◆ Who owns it?

◆ What governance controls apply?

◆ Which SCPs should protect it?

◆ What risks must be isolated?

◆ How will compliance be maintained?

These questions separate cloud administrators from cloud architects.

◈ FINAL TAKEAWAY

AWS Organizations, SCPs, Landing Zones, and Multi-Account Strategies are not simply AWS services.

They are governance mechanisms.

Together they provide:

✔ Security

✔ Scalability

✔ Compliance

✔ Financial Control

✔ Operational Excellence

✔ Enterprise Resilience

The most successful cloud environments are not built through technology alone.

They are built through governance.

MZ UCA PRINCIPLE

"Anyone can create AWS accounts.

Architects design structure.

Governance Architects design control."

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MZ UCA Cloud Academy

Building Architects. Establishing Governance. Creating Leaders.